

P1. 题8 P43

" $\Rightarrow$ ": 若 $\text{grad } u$ 为常向量, 则 $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}$ 均为常数

对 x 积分, $u = ax + \varphi(y, z)$

再对 y 积分, $u = ax + by + \psi(z)$

再对 z 积分, $u = ax + by + cz + d$

" $\Leftarrow$ " $u = ax + by + cz + d$

$\nabla u = (\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}) = (a, b, c)$ 为常向量 $\square$.

P2-3 题8 1, 3

$$I) \begin{cases} x = a \sin^2 t \\ y = b \sin t \cos t \\ z = c \cos^2 t \end{cases} \rightarrow \begin{cases} x' = a \sin 2t \\ y' = b(\cos^2 t - \sin^2 t) = b \cos 2t \\ z' = -c \sin 2t \end{cases}$$

$$M_0 = (\frac{a}{2}, \frac{b}{2}, \frac{c}{2}) \quad \vec{s} = (a, 0, -c)$$

$$\mathcal{L}: \frac{x - \frac{a}{2}}{\frac{a}{2}} = \frac{y - \frac{b}{2}}{0} = \frac{z - \frac{c}{2}}{-c}$$

$$\pi: a(x - \frac{a}{2}) - c(z - \frac{c}{2}) = 0$$

$$III) \vec{r}_1 = (2x-3, 2y, 2z) \rightarrow (-1, 2, 2)$$

$$\vec{r}_2 = (2, -3, 5)$$

$$\vec{s} = \vec{r}_1 \times \vec{r}_2 = (-1, 2, 2) \times (2, -3, 5) = (16, 9, -1)$$

$$\mathcal{L}: \frac{x-1}{16} = \frac{y-1}{9} = \frac{z-1}{-1}$$

$$\pi: 16(x-1) + 9(y-1) - (z-1) = 0$$

P3 习题8 P45

$$\begin{cases} x = e^t \cos t \\ y = e^t \sin t \\ z = e^t \end{cases} \rightarrow \begin{cases} x' = e^t (\cos t - \sin t) \\ y' = e^t (\sin t + \cos t) \\ z' = e^t \end{cases} \quad \vec{S}_1 = (\cos t - \sin t, \sin t + \cos t, 1)$$

$x^2 + y^2 = z^2 \rightarrow$ 曲线在圆锥面上, 母线 $\vec{S}_2 = (\cos t, \sin t, 1)$

$$\theta = \arccos \frac{|\vec{S}_1 \cdot \vec{S}_2|}{|\vec{S}_1| |\vec{S}_2|} = \arccos \frac{|\cos^2 t - \sin t \cos t + \sin t^2 + \sin t \cos t + 1|}{\sqrt{(\cos t - \sin t)^2 + (\sin t + \cos t)^2 + 1} \sqrt{\cos^2 t + \sin^2 t + 1}} = \frac{2}{\sqrt{2} \sqrt{2}} = \arccos \frac{\sqrt{2}}{2}$$

为常数, 故曲线与圆锥面所有母线相交成等角 $\square$.

P4-5 习题8 P46 1, 3

I) $F(x, y, z) = z - \arctan \frac{y}{x}$

$$F_x = \frac{-\frac{y}{x^2}}{1 + (\frac{y}{x})^2} \quad F_y = \frac{\frac{1}{x}}{1 + (\frac{y}{x})^2} \quad F_z = 1$$

$$\vec{n} = (\frac{1}{2}, -\frac{1}{2}, 1)$$

$$\pi: \frac{1}{2}(x-1) - \frac{1}{2}(y-1) + (z-1) = 0$$

$$l: \frac{x-1}{\frac{1}{2}} = \frac{y-1}{-\frac{1}{2}} = \frac{z-1}{1} = 0$$

III) $F(x, y, z) = e^{\frac{x}{z}} + e^{\frac{y}{z}} - 4$

$$F_x = \frac{1}{z} e^{\frac{x}{z}} \quad F_y = \frac{1}{z} e^{\frac{y}{z}} \quad F_z = -\frac{x}{z^2} e^{\frac{x}{z}} - \frac{y}{z^2} e^{\frac{y}{z}}$$

$$\vec{n} = (2, 2, -4 \ln 2) \Rightarrow (1, 1, -2 \ln 2)$$

$$\pi: x - \ln 2 + y - \ln 2 - 2 \ln 2 (z - 1) = 0 \rightarrow x + y - 2 \ln 2 = 0$$

$$l: \frac{x - \ln 2}{1} = \frac{y - \ln 2}{1} = \frac{z - 1}{-2 \ln 2}$$

P₆ 习题8 P₄₇

$$\begin{cases} 4x+y-z-3=0 \\ x+y-z=0 \end{cases}$$

设平面方程: $4x+y-z-3+\lambda(1x+y-z)=0$
 $(4+\lambda)x+(1+\lambda)y-(1+\lambda)z=3$

$$3x^2+y^2-z^2=3 \rightarrow 6xdx+2ydy-2zdz=0 \quad \vec{n}=(3x_0, y_0, -z_0)$$

$$\pi: 3x_0(x-x_0)+y_0(y-y_0)-z_0(z-z_0)=0$$

$$-3x_0^2-y_0^2+z_0^2+3x_0x+y_0y-z_0z=0$$

$$3x_0x+y_0y-z_0z=3$$

$$\begin{cases} 3x_0=4+\lambda \\ y_0=1+\lambda \\ z_0=1+\lambda \\ 3=3 \end{cases} \rightarrow 3\left(\frac{4+\lambda}{3}\right)^2+(1+\lambda)^2-(1+\lambda)^2=3$$

$$\lambda_1=-1 \quad \lambda_2=-7$$

$$\lambda=-1 \quad \pi: x=1 \quad \lambda=-7: \pi: x+2y-2z+1=0$$

P₇ 习题8 P₄₈

$$x^2+y^2+z^2=14$$

$$F_x=2x \quad F_y=2y \quad F_z=2z \rightarrow \vec{n}_1=(-2, -4, 6) \rightarrow (1, 2, -3)$$

$$3x^2+y^2+z^2=16$$

$$F_x=6x \quad F_y=2y \quad F_z=2z \rightarrow \vec{n}_2=(-6, -4, 6) \rightarrow (3, 2, -3)$$

$$\theta = \arccos \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|} = \arccos \frac{3+4+9}{\sqrt{1+4+9} \cdot \sqrt{9+4+9}} = \arccos \frac{16}{11\sqrt{3}}$$

P₈₋₉ 习题8 P₄₉ 2, 3

$$\text{II) } F_x=\frac{2}{3}x^{-\frac{1}{3}} \quad F_y=\frac{2}{3}y^{-\frac{1}{3}} \quad F_z=\frac{2}{3}z^{-\frac{1}{3}}$$

$$\pi: \frac{2}{3}x_0^{-\frac{1}{3}}(x-x_0)+\frac{2}{3}y_0^{-\frac{1}{3}}(y-y_0)+\frac{2}{3}z_0^{-\frac{1}{3}}(z-z_0)=0$$

$$\frac{2}{3}x_0^{-\frac{1}{3}}x+\frac{2}{3}y_0^{-\frac{1}{3}}y+\frac{2}{3}z_0^{-\frac{1}{3}}z-\frac{2}{3}(x_0^{\frac{2}{3}}+y_0^{\frac{2}{3}}+z_0^{\frac{2}{3}})=0$$

$$x_0^{-\frac{1}{3}}x + y_0^{-\frac{1}{3}}y + z_0^{-\frac{1}{3}}z = a^{\frac{2}{3}}$$

截面: $\begin{cases} x_1 = a^{\frac{2}{3}}x_0^{-\frac{1}{3}} \\ y_1 = a^{\frac{2}{3}}y_0^{-\frac{1}{3}} \\ z_1 = a^{\frac{2}{3}}z_0^{-\frac{1}{3}} \end{cases} \quad x_1^2 + y_1^2 + z_1^2 = a^{\frac{4}{3}}(x_0^{\frac{2}{3}} + y_0^{\frac{2}{3}} + z_0^{\frac{2}{3}}) = a^{\frac{4}{3}} \cdot a^{\frac{2}{3}} = a^2 \quad \square$

IV. $z = \pi f(\frac{y}{x})$

$$F(x, y, z) = x f(\frac{y}{x}) - z$$

$$F_x = f(\frac{y}{x}) - \frac{y}{x} f'(\frac{y}{x}) \quad F_y = f'(\frac{y}{x}) \quad F_z = -1$$

$$\pi: [f(\frac{y_0}{x_0}) - \frac{y_0}{x_0} f'(\frac{y_0}{x_0})](x - x_0) + f'(\frac{y_0}{x_0})(y - y_0) - (z - z_0) = 0$$

$$[f(\frac{y_0}{x_0}) - \frac{y_0}{x_0} f'(\frac{y_0}{x_0})]x + f'(\frac{y_0}{x_0})y - z - \underbrace{[f(\frac{y_0}{x_0}) - \frac{y_0}{x_0} f'(\frac{y_0}{x_0})x_0 + f'(\frac{y_0}{x_0})y_0 - z_0]}_0 = 0$$

$$[f(\frac{y_0}{x_0}) - \frac{y_0}{x_0} f'(\frac{y_0}{x_0})]x + f'(\frac{y_0}{x_0})y - z = 0$$

代入 $(0, 0, 0)$ 点, $0 + 0 - 0 = 0$, 成立

故所有切平面都过点 $(0, 0, 0)$ $\square$

P10. 补充 P8

$$F_x = f_1 \cdot (-\frac{y-b}{(\pi-a)^2}) + f_2 \cdot (-\frac{z-c}{(\pi-a)^2}) \quad F_y = f_1 \cdot \frac{1}{\pi-a} \quad F_z = f_2 \cdot \frac{1}{\pi-a}$$

$$\pi: [-f_1(\frac{y_0-b}{\pi-a}) - f_2(\frac{z_0-c}{\pi-a})](x - x_0) + f_1(y - y_0) + f_2(z - z_0) = 0$$

代入 (a, b, c)

$$-f_1(\frac{y_0-b}{\pi-a})(a-x_0) - f_2(\frac{z_0-c}{\pi-a})(x-x_0) + f_1(b-y_0) + f_2(c-z_0)$$

$$= f_1(y_0-b) + f_2(b-y_0) + f_2(z_0-c) + f_2(c-z_0) = 0 \quad \text{成立}$$

故曲面的所有切平面过定点 (a, b, c) $\square$.

P₁₁ 题 8 P₉

$$F(x, y, z) = x^2 + y^2 - z$$

$$F_x = 2x \quad F_y = 2y \quad F_z = -1$$

$$\vec{n} = (2, -4, -1)$$

$$\vec{S} = (1, 1, 0) \times (1, a, -1) = (-1, 1, a-1)$$

$$\vec{n} \cdot \vec{S} = -2 - 4 - a + 1 = 0 \rightarrow a = -5$$

$$\pi: 2(x-1) - 4(y+2) - (z-5) = 0$$

$$2x - 4y - z - 5 = 0$$

$$\text{取 } \pi \text{ 上 } x=0 \text{ 的点 } \begin{cases} y+b=0 \\ -5y-z-3=0 \end{cases} \rightarrow \begin{cases} y=-b \\ z=5b-3 \end{cases}$$

$$4b - 5b + 3 - 5 = 0$$

$$b = -2$$

综上, $a = -5, b = -2$

P₁₂₋₁₃ 题 11 P₄ 1, 4

$$I) \sum_{n=1}^{\infty} \frac{\sin^n x}{n^2}$$

$$\text{由于 } \sin^n x \leq 1, \text{ 故 } \frac{\sin^n x}{n^2} \leq \frac{1}{n^2}$$

由比较法, $\sum_{n=1}^{\infty} \frac{1}{n^2}$ 收敛, 故 $\sum_{n=1}^{\infty} \frac{\sin^n x}{n^2}$ 在 $\mathbb{R}$ 上收敛, 收敛域为 $(-\infty, +\infty)$

$$IV) \sum_{n=1}^{\infty} \frac{1}{n} \left(\frac{1+x}{x} \right)^n$$

$x > 0$, $\sum_{n=1}^{\infty} \frac{1}{n}$ 调和级数不收敛

$x < 0$, $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ 交错级数, $\frac{1}{n+1} < \frac{1}{n}$, $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$, 收敛

故收敛域为 $(-\infty, 0)$

P₁₄₋₁₆ 习题 11 P₁₅ 4, 5, 6

$$IV) \sum_{n=1}^{\infty} \frac{n+1}{2^n} x^{2n-1}$$

偶次项为零

$$\lim_{n \rightarrow \infty} \frac{|u_{n+1}(x)|}{|u_n(x)|} = \frac{n+2}{2(n+1)} x^2 = \frac{1}{2} x^2 < 1 \rightarrow |x| < \sqrt{2}$$

$$x = \sqrt{2} \quad \frac{n+1}{2^n} (\sqrt{2})^{2n-1} = \frac{n+1}{\sqrt{2} 2^{n-1}} (\sqrt{2})^{2n-1} = \frac{n+1}{\sqrt{2}} \quad \lim_{n \rightarrow \infty} u_n(x) \neq 0 \rightarrow \text{不收敛}$$

$$x = -\sqrt{2} \quad \frac{n+1}{2^n} (-\sqrt{2})^{2n-1} = (-1)^{2n-1} \frac{n+1}{\sqrt{2}} \text{ 交错级数 } \lim_{n \rightarrow \infty} \frac{n+1}{\sqrt{2}} \neq 0 \rightarrow \text{不收敛}$$

收敛域为 $(-\sqrt{2}, \sqrt{2})$

$$V) \sum_{n=1}^{\infty} \frac{x^n}{(n+1)^p}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n+2} \right)^p = 1$$

$$x=1 \quad \sum_{n=1}^{\infty} \frac{1}{(n+1)^p} \begin{cases} p > 1 \text{ 收敛} \\ p \leq 1 \text{ 发散} \end{cases}$$

$$x=-1 \quad \sum_{n=1}^{\infty} \frac{(-1)^n}{(n+1)^p} \text{ 交错 } \begin{cases} p \leq 0 \text{ 发散} \\ p > 0 \text{ 收敛} \end{cases}$$

$$\text{收敛域: } \begin{cases} (-1, 1) & , p \leq 0 \\ [-1, 1) & , 0 < p \leq 1 \\ [-1, 1] & , p > 1 \end{cases}$$

$$VI) \sum_{n=1}^{\infty} \frac{n!}{n^n} x^n$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{(n+1)n^n}{(n+1)^{n+1}} = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n = \lim_{n \rightarrow \infty} \left[\left(1 - \frac{1}{n+1} \right)^{-(n+1)} \right]^{-\frac{n}{n+1}} = e^{-1}$$

$$x=e \quad \lim_{n \rightarrow \infty} \frac{n!}{n^n} e^n = \lim_{n \rightarrow \infty} \frac{n^n e^{-n} \sqrt{2\pi n} e^n}{n^n} = \lim_{n \rightarrow \infty} \sqrt{2\pi n} \rightarrow \infty \text{ 发散}$$

$$x=-e \quad (-1)^n \frac{n!}{n^n} e^n \text{ 交错, } \lim_{n \rightarrow \infty} \frac{n!}{n^n} e^n \neq 0 \text{ 发散}$$

收敛域为 $(-e, e)$

P₁₇ 题 11 P₁₇ 6

$$v) \sum_{n=0}^{\infty} \frac{2n+1}{n!} x^{2n}$$

$$= \sum_{n=0}^{\infty} \frac{2n}{n!} x^{2n} + \sum_{n=0}^{\infty} \frac{x^{2n}}{n!}$$

$$= 2 \sum_{n=0}^{\infty} \frac{x^{2n}}{(n-1)!} + e^{x^2}$$

$$= 2x^2 \sum_{n=0}^{\infty} \frac{(x^2)^{n-1}}{(n-1)!} + e^{x^2} = (2x^2)e^{x^2}$$

P₁₈ 题 11 P₁₈ 3

$$\sum_{n=1}^{\infty} \frac{2n+1}{9^n}$$

$$\text{令 } x = \left(\frac{1}{3}\right) \quad S(x) = \sum_{n=1}^{\infty} (2n+1)x^{2n}$$

$$= \sum_{n=1}^{\infty} (x^{2n+1})'$$

$$= \left(\sum_{n=1}^{\infty} x^{2n+1} \right)' = \left(\frac{x^3}{1-x^2} \right)' = \frac{-x^4 + 3x^2}{(1-x)^2}$$

$$\sum_{n=1}^{\infty} \frac{2n+1}{9^n} = \frac{-\frac{1}{81} + \frac{1}{9}}{\left(\frac{8}{9}\right)^2} = \frac{13}{32}$$

P₁₉₋₂₀ 题 11 P₁₉ 2, 4

$$II) \frac{1}{x^2}$$

$$\text{令 } t = x-1 \rightarrow x = t+1$$

$$\frac{1}{t+1} = \sum_{n=0}^{\infty} (-1)^n t^n \quad (-1 < t < 1)$$

$$-\frac{1}{(t+1)^2} = \sum_{n=1}^{\infty} (-1)^n \cdot n t^{n-1} \rightarrow \frac{1}{(t+1)^2} = \sum_{n=1}^{\infty} (-1)^{n+1} n t^{n-1}$$

$$= \sum_{n=0}^{\infty} (-1)^n (n+1) t^n$$

$$\frac{1}{x^2} = \sum_{n=0}^{\infty} (-1)^n (n+1) (x-1)^n \quad x \in (0, 2)$$

$$IV) \ln(10+\pi), \pi_0=0$$

$$= \ln(10(1+\frac{\pi}{10})) = \ln 10 + \ln(1+\frac{\pi}{10})$$

$$= \ln 10 + \sum_{n=1}^{\infty} (-1)^{n-1} \frac{(\frac{\pi}{10})^n}{n} \quad (-10 < \pi \leq 10)$$

思考题

$$f(x,y) = \begin{cases} \frac{y^3}{x^2+y^2} & , (x,y) \neq (0,0) \\ 0 & , (x,y) = (0,0) \end{cases}$$

$$I) 0 \leq \left| \frac{y^3}{x^2+y^2} \right| = \left| \frac{y^2}{x^2+y^2} \cdot y \right| \leq |y|$$

由夹逼定理, $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0 = f(0,0) \rightarrow$ 连续 $\square$.

$$II) \frac{\partial f}{\partial x} \Big|_{(0,0)} = \lim_{t \rightarrow 0} \frac{f(t \cos \alpha, t \cos \beta) - f(0,0)}{t} = \lim_{t \rightarrow 0} \frac{\frac{t^3 \cos^3 \beta}{t^2} - 0}{t} = \cos^3 \beta$$

按任意方向向上方向导数存在, 为 $\cos^3 \beta$ $\square$.

$$III) \frac{\partial f}{\partial x} \Big|_{(0,0)} = \lim_{\alpha \rightarrow 0} \frac{f(0, \cos \alpha) - f(0,0)}{\cos \alpha} = \lim_{\alpha \rightarrow 0} \frac{0-0}{\cos \alpha} = 0, \quad \frac{\partial f}{\partial y} \Big|_{(0,0)} = \lim_{\beta \rightarrow 0} \frac{f(0, \cos \beta) - f(0,0)}{\cos \beta} = \lim_{\beta \rightarrow 0} \frac{\cos^3 \beta}{\cos \beta} = 1$$

$$\text{若可微, 则 } \frac{\partial f}{\partial x} \Big|_{(0,0)} = \frac{\partial f}{\partial x} \cos \alpha + \frac{\partial f}{\partial y} \cos \beta = \cos^3 \beta \neq \cos^3 \beta$$

故不可微 $\square$